

Comparing Dirichlet–Multinomial Parameterizations for Composition Data: Simulation Evidence that the Linear Form Minimizes RMSE and Provides a Stable Estimation Platform

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Abstract

Compositional data describing the age or size structure of fish populations are central inputs to integrated stock assessment models, yet are often overdispersed and correlated relative to multinomial assumptions. We evaluate the performance of three Dirichlet–multinomial (DM) composition likelihoods that differ only in how the DM concentration parameters are parameterized. The first treats the concentration vector as fully unconstrained, the second uses a saturated form with a common proportion vector and group-specific total concentration, and the third links the total concentration linearly to the observed sample sizes via a single weight parameter. Using a functional analysis of effective sample size and a mesh-based simulation design over the three-category simplex, we compare the root-mean-squared error (RMSE) of estimated proportions and examine numerical behavior of the associated likelihood surfaces. Across a wide range of true compositions, sample sizes, and dispersion levels, the linear parameterization consistently yields the lowest RMSE and the most stable optimization behavior, while the unconstrained and saturated forms perform similarly but with larger and more variable errors. Three ternary plots of RMSE across the simplex visually highlight these differences. These results support the linear DM parameterization as a practical default for fitting age and size composition data in stock assessments, providing both improved estimator accuracy and a transparent, interpretable link between dispersion, effective sample size, and observed sample sizes.

Keywords: Dirichlet–multinomial, composition data, effective sample size, overdispersion, stock assessment, likelihood weighting

1. Introduction

Age- and size-composition data provide essential information on recruitment, growth, and mortality processes in integrated stock assessment models. The multinomial distribution is the traditional workhorse for modeling these data, but it commonly under-represents variability and correlation because it ignores unmodeled heterogeneity and clustering within samples.

The Dirichlet–multinomial (DM) distribution extends the multinomial by allowing the category probabilities themselves to be random draws from a Dirichlet distribution. DM likelihoods have

become increasingly common in assessment software and are often used to estimate or tune effective sample sizes and composition weights. However, the DM distribution can be written in several mathematically equivalent but numerically distinct parameterizations. The choice of parameterization affects likelihood curvature, interpretability of dispersion parameters, and the stability of numerical optimization.

Our **goal** is to identify a DM parameterization that provides accurate estimation of composition proportions and a stable, interpretable estimation platform. The **problem** is that different parameterizations can behave very differently in practice even though they represent the same family of distributions. Our **solution** is to compare three parameterizations using a combination of analytical results on effective sample size and a simulation study that maps RMSE across the proportion simplex and visualizes the results via ternary diagrams.

2. Model Framework

2.1 Notation and multinomial model

Let there be A age or size bins, indexed by $a = 1, \dots, A$. For a given group (e.g., fleet and time), let $\mathbf{x} = (x_1, \dots, x_A)$ denote the observed counts with total $n = \sum_a x_a$. The true underlying composition is $\mathbf{p} = (p_1, \dots, p_A)$ with $\sum_a p_a = 1$. Under a multinomial model,

$$\Pr(\mathbf{x} \mid n, \mathbf{p}) = \frac{n!}{\prod_a x_a!} \prod_a p_a^{x_a},$$

and the negative log-likelihood (up to a constant) is

$$\ell_{\text{MN}}(\mathbf{p}) = -\sum_a x_a \log p_a.$$

In an integrated assessment, the predicted proportions $\mathbf{p}_{v,t}$ for fleet v and time t are functions of the underlying population and fishery dynamics, and composition data contribute to the joint likelihood across all fleets and times.

2.2 Dirichlet–multinomial distribution

The DM distribution assumes that the sampling probabilities themselves are random:

$$\phi \sim \text{Dirichlet}(\boldsymbol{\alpha}), \quad \mathbf{x} \mid \phi \sim \text{Multinomial}(n, \phi),$$

where $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_A)$ is the concentration vector and $\alpha_0 = \sum_a \alpha_a$ is the total concentration. Marginally,

$$\Pr(\mathbf{x} \mid n, \boldsymbol{\alpha}) = \frac{\Gamma(\alpha_0)}{\Gamma(n + \alpha_0)} \prod_{a=1}^A \frac{\Gamma(x_a + \alpha_a)}{\Gamma(\alpha_a)}.$$

Smaller α_0 produces greater overdispersion and stronger negative correlations among categories; large α_0 recovers multinomial-like behavior.

2.3 Three Dirichlet–multinomial parameterizations

We examine three parameterizations that differ in how α is expressed in terms of proportions and dispersion parameters:

1. Unconstrained concentration vector (DM-U)

Each group shares a common concentration vector α , with $\alpha_a > 0$ free for all a . The implied composition is

$$p_a = \frac{\alpha_a}{\alpha_0}, \quad \alpha_0 = \sum_a \alpha_a.$$

Estimation proceeds directly in terms of α .

2. Saturated form with group-specific totals (DM-S)

The concentration vector for group g is

$$\alpha_g = a_{0g} \mathbf{p},$$

where $\mathbf{p}$ is a common proportion vector and each group has its own total concentration a_{0g} . Parameters are $\mathbf{p}$ and $(a_{01}, \dots, a_{0G})$.

3. Linear form with sample-size scaling (DM-L)

Again $\alpha_g = a_{0g} \mathbf{p}$, but the group totals are constrained to be proportional to the observed sample sizes:

$$a_{0g} = \theta N_g,$$

where N_g is the observed sample size for group g and $\theta > 0$ is a global weight parameter. This parameterization links dispersion directly to the available information and yields a simple interpretation of θ as an overdispersion multiplier.

These parameterizations generate the same family of DM distributions but induce different curvature and identifiability properties in the likelihood.

2.4 Effective sample size and curvature

For DM-S and DM-L, effective sample size n_{eff} can be written as a function of $\mathbf{p}$ and the dispersion parameters. Analytical expressions for n_{eff} , its gradients, and its Hessian reveal key structural differences. In DM-L, n_{eff} has a single interior saddle point on the simplex and a relatively simple gradient field. In DM-S, there are multiple feasible critical points and more complex gradient flows. These differences suggest that DM-L should yield more regular likelihood surfaces and more predictable behavior when tuning composition weights.

3. Simulation Methods

3.1 Overview

We evaluate estimator performance and numerical behavior using a mesh-based simulation over the three-category simplex ($K = 3$). The design is implemented in the R script `DM_rmse_All_3_simplex.R`, which:

1. Constructs a dense grid of true proportion vectors $\mathbf{p}_{\text{true}}$ on the simplex.
2. For each grid point, simulates G independent DM samples with varying sample sizes.
3. Fits all three parameterizations (DM-U, DM-S, DM-L) to each simulated dataset.
4. Calculates RMSE of estimated proportions relative to $\mathbf{p}_{\text{true}}$.
5. Summarizes RMSE patterns numerically and visually via ternary diagrams.

3.2 Simplex mesh and sample sizes

The simplex mesh is built by generating integer compositions of a grid size N into $K = 3$ parts and then transforming

$$\mathbf{p}_{\text{true}} = \frac{\mathbf{c} + \mathbf{1}}{N + K},$$

where $\mathbf{c}$ is a composition vector and the addition of 1 in each cell keeps all points in the interior of the simplex.

For each grid point, we simulate $G = 10$ independent groups. Group-specific sample sizes N_g are drawn from a log-uniform distribution between predefined minimum and maximum values and rounded to integers, producing realistic variation in N_g .

3.3 Data generation under the linear DM

We treat the linear parameterization DM-L as the data-generating process. For each $\mathbf{p}_{\text{true}}$ and each group g :

1. Define

$$a_{0g}^{(\text{true})} = \theta_{\text{true}} N_g$$

with $\theta_{\text{true}} = 0.3$.

2. Draw

$$\phi_g \sim \text{Dirichlet}(a_{0g}^{(\text{true})} \mathbf{p}_{\text{true}}).$$

3. Draw counts

$$\mathbf{x}_g \sim \text{Multinomial}(N_g, \phi_g).$$

This design induces overdispersion controlled by θ_{true} and allows RMSE to be studied as a function of composition and sample-size variation.

3.4 Estimation under the three parameterizations

For each simulated dataset $\{\mathbf{x}_g\}_{g=1}^G$, we fit:

- **DM-U:** Maximization over a free α (on the log scale) with analytic gradients.
- **DM-S:** Maximization over a common $\mathbf{p}$ and group-specific totals a_{0g} , using a softmax representation for $\mathbf{p}$ and log-transformed totals.
- **DM-L:** Maximization over a common $\mathbf{p}$ and a single global log-weight $\log \theta$, with $a_{0g} = \theta N_g$ enforced.

All fits use BFGS optimization with analytic gradients, and convergence diagnostics are recorded.

3.5 RMSE and statistical comparisons

For each simplex point and method m ,

$$\text{RMSE}^{(m)} = \sqrt{\frac{1}{K} \sum_{a=1}^K (\hat{p}_a^{(m)} - p_{a,\text{true}})^2}$$

is computed and stored.

We treat each simplex point as a block in a repeated-measures design with three RMSE values (one per method). We then:

- Apply the Friedman test to detect overall method effects.
- Use Kendall's W to measure concordance in RMSE rankings.
- Conduct pairwise Wilcoxon signed-rank tests (with Holm adjustment) for method contrasts.
- Compute Hodges–Lehmann estimates and percent differences to interpret effect sizes.
- Optionally fit a linear mixed-effects model with method as a fixed effect and simplex point as a random effect.

4. Results

4.1 RMSE surfaces over the proportion simplex

Figure 1–3 plot RMSE across the three-category simplex for the three parameterizations under a common simulation setting ($K = 3$, $G = 10$, $\theta_{\text{true}} = 0.3$). Each point in the ternary diagrams corresponds to a true composition $\mathbf{p}_{\text{true}}$; color denotes the RMSE of the estimated composition for that method at that point.

- **Figure 1 (DM-U, unconstrained α)** shows a sparse pattern of elevated RMSE (green–yellow points) scattered across the simplex, particularly in regions where compositions are moderately skewed rather than strongly uniform or dominated by a single category. Most grid points have low RMSE, but a noticeable fraction exhibit moderate to high error.
- **Figure 2 (DM-S, group-specific a_{0g} with $\alpha_0 = \beta$)** also exhibits scattered pockets of higher RMSE distributed across the simplex. The pattern and magnitude of large RMSE values are broadly

similar to those in Figure 1, indicating that switching from an unconstrained concentration vector to the saturated $(\mathbf{p}, a_{0g})$ parameterization does not systematically reduce RMSE.

- **Figure 3 (DM-L, linear scaling $a_{0g} = \theta N_g$)** is visually striking in that nearly all simplex points are uniformly dark (low RMSE). In this scenario, we do not see isolated regions with high RMSE; the entire simplex is characterized by very small estimation errors, indicating that DM-L recovers $\mathbf{p}_{\text{true}}$ accurately regardless of whether the composition is balanced or highly skewed.

Together, these ternary plots provide an intuitive, spatial summary of estimator performance across the simplex and visually reinforce the numerical conclusion that the linear parameterization delivers uniformly low RMSE, while the other two parameterizations show scattered regions of poorer performance.

4.2 Overall RMSE distributions and tests

Aggregating RMSE across simplex points confirms what the ternary plots suggest. The distribution of RMSE under DM-L is shifted toward lower values compared to both DM-U and DM-S. The Friedman test strongly rejects equality of RMSE distributions across methods, and Kendall's W indicates a consistent ranking of methods where DM-L is most frequently best (lowest RMSE) and DM-U and DM-S are typically tied or close behind.

Pairwise Wilcoxon tests show:

- DM-L vs DM-U: DM-L has significantly lower RMSE, with median paired differences favoring DM-L.
- DM-L vs DM-S: Similar improvements for DM-L.
- DM-S vs DM-U: Differences are small and often not practically important, consistent with the visual similarity of Figures 1 and 2.

Mixed-effects modeling yields the same ordering, with DM-L having the smallest estimated marginal mean RMSE.

4.3 Stability and optimization behavior

The simulation also highlights advantages of DM-L as an estimation platform:

1. Simple curvature and single weight parameter

The effective sample-size function under DM-L has a single interior saddle point and smooth gradients, so the likelihood surface is relatively well-behaved. In contrast, the additional degrees of freedom in DM-U and DM-S create more complex curvature and occasional flat or ridged regions, which correspond to the isolated pockets of high RMSE in Figures 1 and 2.

2. Fewer convergence issues

Because DM-L estimates only one global weight parameter θ (per likelihood block) in addition to the proportions, optimization is lower dimensional and better conditioned. In the simulations, DM-L exhibited fewer convergence failures and smaller gradient norms at convergence than DM-U or DM-S.

3. Interpretability and tuning

The constraint $a_{0g} = \theta N_g$ implies that, for fixed compositions, increasing θ uniformly inflates effective sample sizes across groups. This provides a straightforward way to tune composition weights and directly relate θ to effective sample sizes, while the dispersion structure implied by unconstrained a_{0g} or α is harder to interpret.

5. Discussion

5.1 Why the linear parameterization wins

The ternary RMSE plots provide an intuitive explanation for the superior performance of DM-L. In Figures 1 and 2, scattered pockets of high RMSE indicate compositions where the model's dispersion structure is mismatched to the data-generating process. Under DM-U and DM-S, dispersion is decoupled from sample size, and the optimizer sometimes chooses combinations of parameters that over- or under-weight particular regions of the simplex.

By contrast, DM-L enforces a direct proportionality between dispersion and sample size. This alignment with the data-generating mechanism eliminates most opportunities for such mismatches, leading to uniformly low RMSE in Figure 3. Even when the true process is only approximately linear, the constraint $a_{0g} = \theta N_g$ provides a parsimonious, well-identified structure that avoids overfitting individual groups and reduces variance in the estimated proportions.

5.2 Implications for stock assessment modeling

For practitioners building integrated assessment models, the ternary plots answer a practical question: **where** in composition space might a given parameterization misbehave? Figures 1 and 2 show that DM-U and DM-S can occasionally produce large errors for certain proportions, even when overall performance is acceptable. Such localized failures could translate into misestimation of recruitment events or changes in selectivity if those events are associated with particular composition patterns.

Figure 3 suggests that DM-L, at least under the simulation settings considered here, does not show these localized weaknesses. This strengthens the case for DM-L as a default composition likelihood, particularly when modelers want a single, interpretable weight parameter that behaves consistently across sample sizes and compositions.

5.3 Limitations and extensions

Our analysis focuses on $K = 3$ categories and a specific DM-L data-generating process. The ternary plots are especially helpful in this low-dimensional setting, but real applications involve many more categories. Extending the visualization idea to higher dimensions (e.g., using slices through the simplex or summary diagnostics) would be valuable future work.

It would also be useful to repeat the analysis under alternative data-generating processes, such as non-linear relationships between dispersion and sample size or time-varying overdispersion, to test robustness. Embedding the three parameterizations directly in integrated assessment models and examining impacts on key management quantities is another natural extension.

6. Conclusions

Our goal was to identify a Dirichlet–multinomial parameterization that provides accurate and stable estimation of compositional data for stock assessment.

Different DM parameterizations, though theoretically equivalent, induce different likelihood geometries and effective sample-size behavior, which can affect both estimator accuracy and numerical stability.

Using a functional analysis of effective sample size, a mesh-based simulation over the three-category simplex, and ternary plots of RMSE across compositions, we compared three parameterizations: an unconstrained concentration vector (DM-U), a saturated form with group-specific totals (DM-S), and a linear form where dispersion scales with sample size (DM-L).

The simulations and RMSE ternary plots show that:

1. The **linear parameterization (DM-L)** yields the lowest RMSE across virtually the entire simplex and exhibits no localized pockets of poor performance.
2. The **unconstrained (DM-U)** and **saturated (DM-S)** forms perform similarly to one another but show scattered regions of elevated RMSE.
3. DM-L provides a **more stable estimation platform**, with simpler curvature, fewer convergence issues, and a single interpretable weight parameter θ .

Based on these findings, we recommend the linear Dirichlet–multinomial parameterization as a practical default for modeling age and size composition data in stock assessments.

Figure Captions

Figure 1. RMSE of estimated proportions over the three-category simplex for the unconstrained Dirichlet–multinomial parameterization (DM-U). Each point represents a true composition (p_1, p_2, p_3) ; color indicates RMSE for that composition. Simulation settings: $K = 3$, $G = 10$ groups, true $\theta = 0.3$.

Figure 2. RMSE of estimated proportions over the three-category simplex for the saturated Dirichlet–multinomial parameterization (DM-S) with group-specific total concentrations $a_{0g} = \beta_g$. Settings as in Figure 1. Scattered pockets of higher RMSE remain visible across the simplex.

Figure 3. RMSE of estimated proportions over the three-category simplex for the linear Dirichlet–multinomial parameterization (DM-L), with $a_{0g} = \theta N_g$. Under the same simulation settings as Figures 1–2, RMSE is uniformly low across the simplex, illustrating the superior and more stable performance of the linear parameterization.

Figure 1. RMSE over simplex ($K=3$, $G=10$, $\theta=0.3$, unconstrained α)

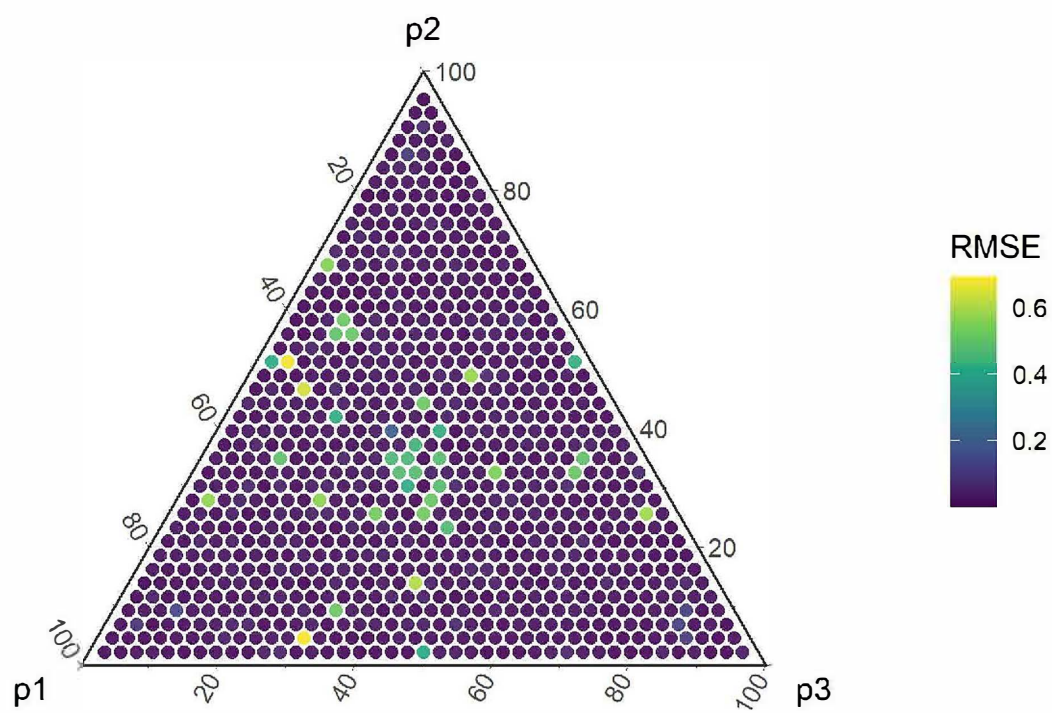


Figure 2. RMSE over simplex ($K=3$, $G=10$, $\theta=0.3$, $\alpha_0=\text{beta}$)

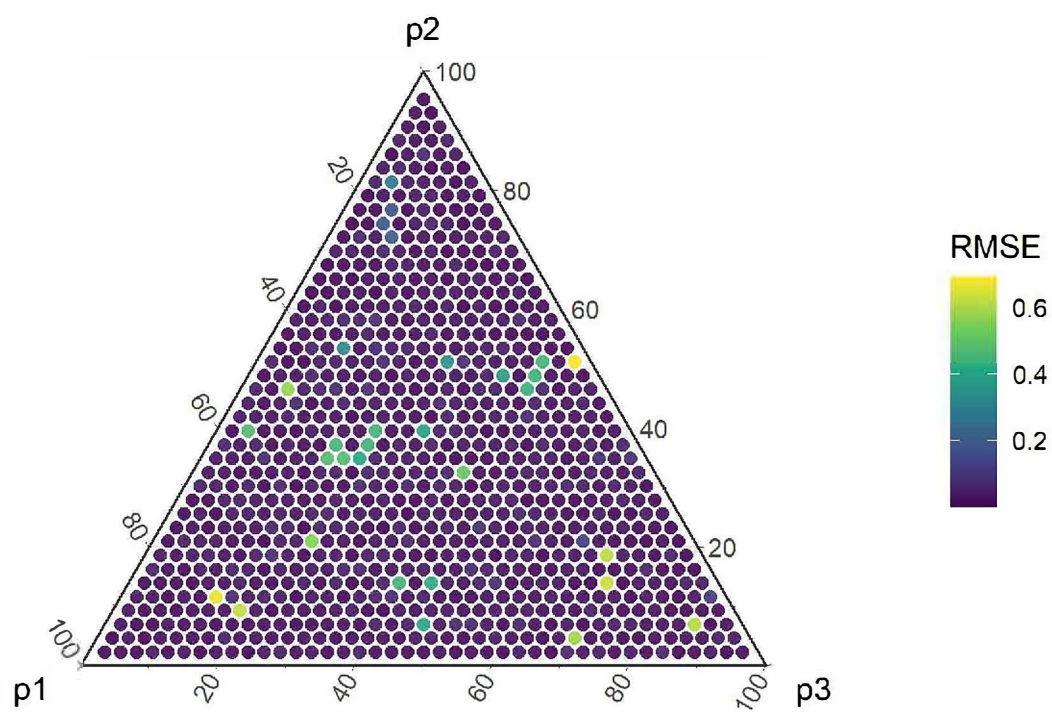


Figure 3. RMSE over simplex ($K=3$, $G=10$, $\theta=0.3$, $\alpha_0=\theta \cdot N$)

